

# ECE 447

## Lesson 22

### Digital

### Communication and Line Coding

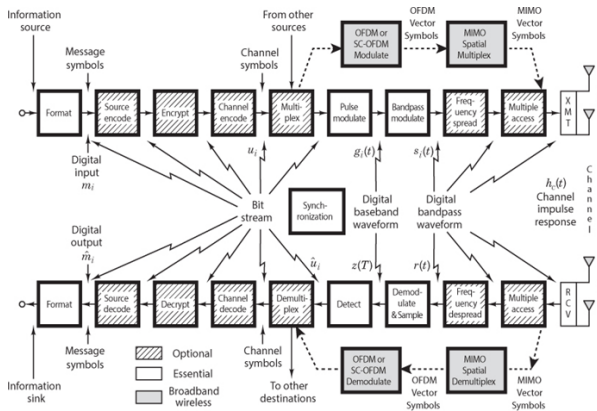
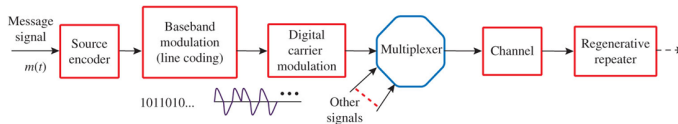


UNITED STATES  
AIR FORCE  
ACADEMY

# SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
  - **HW3.** Grading...
  - **HW4.** Posted on course website. Due Lesson 23 (Tue, 13 Oct).
  - **Lab 5.** Due Lesson 24 (Thu, 15 Oct).

# DIGITAL COMMUNICATION SYSTEMS

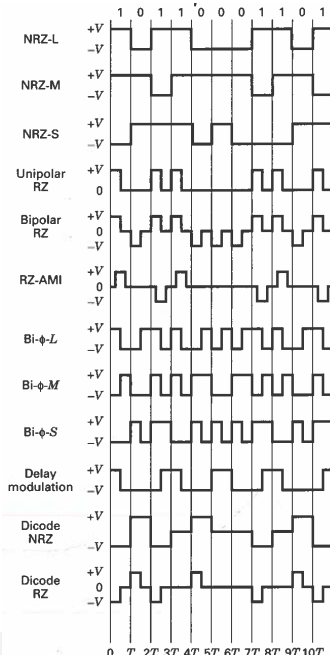
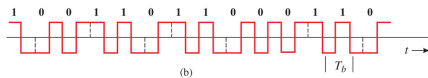
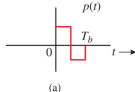
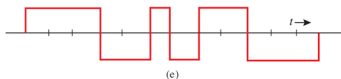
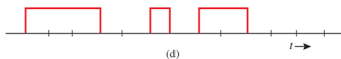
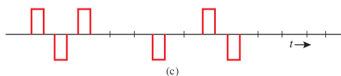
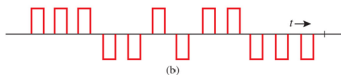
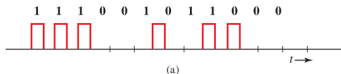


# BASEBAND LINE CODING

Desired line code properties:

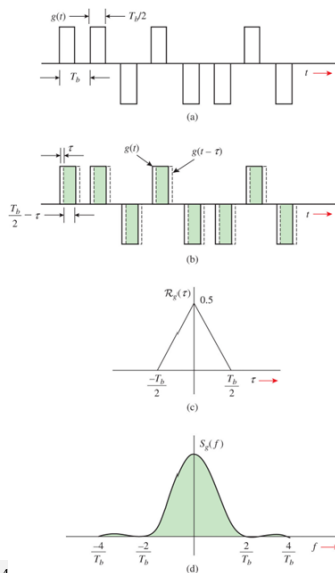
- Low bandwidth
- Power efficiency - low tx power for given bandwidth and error detection rate
- Error detection/correction capability
- No DC content
- Timing info (self-clocking)
- Transparency

# TYPES OF LINE CODES

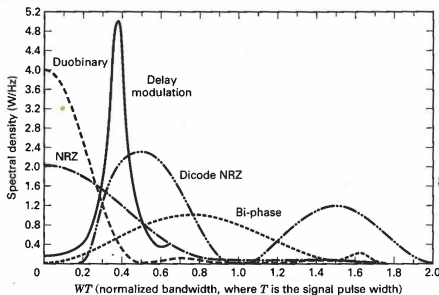
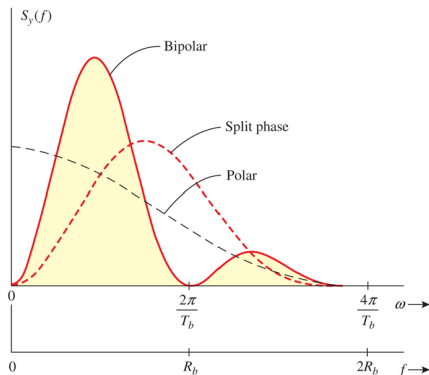


# LINE CODE PSDs

## Example 3.24 - Our good friend, Autocorrelation!



# LINE CODE SPECTRA



**Figure 2.23** Spectral densities of various PCM waveforms.

# LINE CODE TO BASEBAND WAVEFORM

